
RGDA-Tokenizer: Retrieval-Guided Data-Assimilated Tokenizer for Video Continuation

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Abstract

Text-image-to-video continuation models such as Wan2.2 TI2V generate future motion from a text prompt and a single anchor image, which leaves a structural gap between the model’s conditioning space and the richer temporal evidence available in retrieved real videos. This paper reformulates that gap as a data-assimilation problem. Given the first half of a short clip, the frozen TI2V backbone first produces its native prompt-and-image conditioning token for later-half generation. An external retrieval bank then provides top- K real-video observations, which are projected into the TI2V conditioning space through an anchor-image summary, a depth summary, and a motion summary. A small trainable gain network fuses the native conditioning token and the observation innovation to produce a *data-assimilated conditioning token*, which is fed into the frozen Wan DiT for later-half generation. We describe the resulting data-assimilated tokenizer, derive its frozen-backbone diffusion training objective, and specify a concrete training plan with OpenVid-1M as the retrieval corpus and MSR-VTT as the supervised continuation benchmark. The paper is intended as a clean preliminary formulation and implementation plan for retrieval-guided conditioning-space data assimilation in video continuation.

1 Introduction

Modern video world models are increasingly deployed as latent simulators for planning, animation, and long-form generation, but their continuation mechanism is often surprisingly narrow. In Wan2.2 TI2V, the model predicts future video from a text prompt and an anchor image; in practice, that anchor is commonly the last frame from the previous clip. This interface is elegant, but it also creates a clear bottleneck: the model must infer the entire future temporal evolution from a single image token, even though related real videos may provide much richer motion and geometry evidence.

This paper argues that the missing component is not necessarily a larger prior, but a *conditioning-space analysis update*. In classical data assimilation (DA), one starts from a prior state, compares it against external observations, and then computes an analysis state that better reflects the available evidence. We transfer that logic to TI2V continuation with one important clarification: the prior is *not* a separately predicted future video. Instead, the first half of a 10-second clip defines the continuation context; the frozen TI2V prior forms its native prompt-and-image conditioning token; retrieved real videos act as an external observation bank; and a small gain network produces a data-assimilated continuation condition that replaces the raw conditioning token before the later-half rollout.

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The difficult step is not retrieval itself. The real difficulty is that the retrieved observation is a *video*, while TI2V conditions on an *image-plus-text* conditioning space. We therefore compress each retrieved clip into an anchor-image observation and a compact temporal observation consisting of depth and motion summaries. This converts the heterogeneous retrieved video into a structured observation that can be compared with the native TI2V condition in a common space. The resulting method preserves the DA interpretation:

1. form the native prompt-and-image conditioning token from the frozen TI2V prior,
2. retrieve observation videos from an external corpus,
3. project those videos into a shared observation space,
4. compute an innovation,
5. produce a data-assimilated conditioning token with a single trainable gain network.

The goal of the current draft is to make this formulation concrete enough to implement. We therefore focus on the conditioning-token design, the training objective, the dataset split between retrieval and supervised learning, and a practical training loop rather than claiming final benchmark results.

2 Related Work

World models and long-horizon rollout. Latent world models compress observations into a state representation and then evolve that representation forward with a learned dynamics model. This design underlies both classical latent control architectures and modern generative world models. Ha and Schmidhuber [5] made the paradigm explicit by separating representation, memory, and control, while recent surveys argue that modern video generators are increasingly being interpreted through a world-model lens [6, 17]. A large 2025–2026 line of work attacks long-horizon drift by strengthening state persistence, memory, or context. Examples include retrieval- or state-conditioned video world models [3, 18], compressed or explicit spatial memory [11, 16], longer-horizon embodied rollouts with mixture-of-experts or closed-loop training [14, 21], and persistent or embodiment-aware latent states [23, 4]. These works demonstrate that better state representations and memory mechanisms are critical, but they still largely handle external evidence through conditioning, retrieval augmentation, or latent persistence rather than through an explicit prior–observation–analysis state-correction step.

Retrieval, external memory, and observation sources. Retrieval and external memory have become common tools for improving video realism and long-horizon coherence. Retrieval-augmented generation appears in systems such as Peruzzo et al. [12] and Chen et al. [3], while explicit memory banks or persistent scene stores are central to Oshima et al. [11], Wang et al. [16], and Wu et al. [18]. In most of these methods, retrieved evidence is used as conditioning context, as a state prompt, or as an auxiliary memory. Our formulation is deliberately narrower and more explicit: retrieved evidence is converted into a structured observation, weighted through a retrieval-dependent operator, and assimilated into the latent state estimate. This distinction is essential because it turns retrieval from a conditioning heuristic into a measurable source of new information.

Multimodal retrieval and dataset support. Our training plan depends on an observation bank that is both prompt-annotated and large enough to supply semantically related real clips. OpenVid-1M is one of the strongest current choices for this role because it provides over one million text-video pairs and a curated high-definition subset [10]. For retrieval and prompt-alignment scoring we use LanguageBind, which places text, image, video, and depth features into a shared representation space [24]. For supervised token learning on fixed-length clips, MSR-VTT is a practical complementary benchmark because it provides 10K caption-annotated web videos and a widely used public train/test split [19].

Data assimilation with learned and generative priors. DA provides a principled answer to sequential state estimation under model error and partial observations. The broader DA literature has already argued that latent and reduced-order spaces are natural places to fuse machine learning with estimation [1]. More recent work combines DA with learned priors or generative models, including sparse-observation weather DA [9], derivative-free ensemble guidance for diffusion models [22],

stochastic-interpolant DA [2], and diffusion or flow-based perspectives on posterior correction and sequential warm starts [7, 13, 8]. These papers show that DA-style correction remains useful even when the predictive prior is highly expressive. However, they are not focused on world models with retrieved visual memory, geometric observation operators, or long-horizon video drift. That gap is where our method sits.

Geometry-aware observations. To make retrieved evidence useful for state estimation, the observation should be structured rather than purely raw RGB. Monocular geometry estimators such as Depth Anything V2 [20] offer a practical route: convert retrieved real clips into depth-aware observations that describe scene layout and coarse 3D consistency. In our setting, depth is not treated as full physical supervision; rather, it supplies a stable geometry summary that complements the anchor image and motion summaries when projecting retrieved videos into TI2V token space.

3 Preliminaries

3.1 TI2V Continuation Setting

Let a supervised clip be a prompt-annotated 10-second video

$$x = (x_1, \dots, x_T), \quad p \in \mathcal{P}, \quad (1)$$

with midpoint split

$$x^{\text{ctx}} = x_{1:T/2}, \quad x^{\text{tgt}} = x_{T/2+1:T}. \quad (2)$$

In Wan2.2 TI2V [15], the later-half continuation is conditioned primarily on the last available image and the text prompt. We therefore define

$$I_{\text{last}} = x_{T/2}, \quad (3)$$

and let the frozen TI2V context encoder produce the native continuation-conditioning token

$$c^0 = C_{\text{wan}}(I_{\text{last}}, p). \quad (4)$$

The target future clip is still encoded into latent space through the frozen Wan tokenizer or VAE:

$$z^{\text{tgt}} = E_{\text{wan}}(x^{\text{tgt}}). \quad (5)$$

The token c^0 is the *base conditioning token*: the native TI2V condition computed directly from the prompt and last context frame before any data-assimilation update. The latent z^{tgt} is the supervised target latent for the actual later half.

3.2 Retrieval Memory and Shared Embedding Space

We assume a retrieval bank

$$\mathcal{R} = \{(r_i, c_i)\}_{i=1}^N, \quad (6)$$

where r_i is a real video clip and c_i is its caption. A LanguageBind encoder [24] maps text, image, and video inputs into a shared embedding space. Given the prompt and the last context frame, we build the retrieval query

$$q = Q_{\text{LB}}(p, I_{\text{last}}), \quad (7)$$

retrieve the top- K clips, and compute normalized weights

$$\alpha_k = \frac{\exp(s(q, r_k)/\tau)}{\sum_{j=1}^K \exp(s(q, r_j)/\tau)}. \quad (8)$$

3.3 Observation Summaries

The retrieved observation is a video, but the TI2V conditioning space is centered on image-plus-text inputs. To bridge that mismatch, we summarize each retrieved clip with three components:

1. an *anchor-image observation* for appearance and scene identity,
2. a *depth observation* for geometry,
3. a *motion observation* for temporal evolution.

Let $\Psi(\cdot)$ be a frozen depth extractor such as Depth Anything V2, and let $\Phi(\cdot)$ be a frozen motion extractor such as dense optical flow. These operators define the observation space that will later be compared with the base conditioning token.

4 Method

4.1 Overview

Our method trains exactly one new module: a *data-assimilated tokenizer* or *gain network* G_ϕ . Everything else remains frozen, including Wan2.2 TI2V, LanguageBind, Depth Anything V2, and the retrieval index. The method is organized into four steps:

1. obtain the native TI2V conditioning token from the prompt and context image;
2. retrieve top- K real clips and project them into a shared observation summary;
3. compute the innovation between the conditioning-side summary and the observation summary;
4. predict a token correction that turns the native conditioning token into a data-assimilated continuation token.

4.2 Base Conditioning Token from the Frozen TI2V Prior

Although the supervised training clip provides the full first-half context, the frozen TI2V model only receives the same conditioning interface it will see at inference time: the last context frame and the text prompt. Let

$$c^0 = C_{\text{wan}}(I_{\text{last}}, p) \quad (9)$$

be the native conditioning token generated by the frozen Wan2.2 TI2V prompt-and-image pathway. Importantly, c^0 is *not* obtained by first generating a future clip and then re-encoding that clip. It is the DiT input condition itself, before any data-assimilation correction is applied.

4.3 Projecting Retrieved Videos into TI2V Observation Space

The retrieved observation is a video with nontrivial temporal structure, while TI2V expects image-centered conditioning. We therefore summarize each retrieved clip r_k into an anchor-image token, a depth token, and a motion token.

We first choose the anchor frame

$$\tilde{I}_k = r_{k,u^*}, \quad u^* = \arg \max_u \cos(E_{\text{img}}(r_{k,u}), E_{\text{img}}(I_{\text{last}})), \quad (10)$$

which selects the retrieved frame most compatible with the current context boundary.

We then define the per-clip observation summary

$$y_k^o = \text{concat}(E_{\text{img}}(\tilde{I}_k), E_{\text{depth}}(\Psi(r_k)), E_{\text{mot}}(\Phi(r_k))), \quad (11)$$

where E_{img} , E_{depth} , and E_{mot} are frozen summary encoders on top of LanguageBind image features, depth features, and motion features respectively. The retrieved observation is the weighted average

$$y^o = \sum_{k=1}^K \alpha_k y_k^o. \quad (12)$$

The native TI2V conditioning token is projected into the same space through a frozen summary operator

$$y^0 = H(c^0). \quad (13)$$

The resulting innovation is

$$r = y^o - y^0. \quad (14)$$

This is the key DA quantity: it measures how the retrieved observation disagrees with the native TI2V condition in a common observation space even though the original modalities differ.

4.4 Data-Assimilated Tokenizer

Let

$$e_p = E_{\text{text}}(p), \quad e_I = E_{\text{img}}(I_{\text{last}}) \quad (15)$$

denote the prompt embedding and the last-frame embedding. The only trainable module is the gain network

$$\Delta c = G_\phi(\text{concat}(c^0, y^0, y^o, r, e_p, e_I)), \quad (16)$$

which predicts a token correction with the same dimensionality as the conditioning token. The final data-assimilated token is

$$c^a = c^0 + \Delta c. \quad (17)$$

At inference time, c^a replaces c^0 as the later-half continuation condition that is fed into the frozen Wan DiT tail for second-half generation.

4.5 Training Objective

Because the gain network modifies the DiT input condition rather than directly predicting a target video token, the natural objective is the frozen Wan diffusion likelihood on the ground-truth later half. Let

$$z_t^{\text{noisy}} = \alpha_t z_t^{\text{tgt}} + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (18)$$

denote the standard noised target latent used by the frozen Wan diffusion objective. The gain network is trained through the frozen denoiser

$$\hat{\epsilon} = \epsilon_\theta(z_t^{\text{noisy}}, t \mid c^a), \quad (19)$$

and optimized with the usual denoising surrogate

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t, \epsilon} [\|\hat{\epsilon} - \epsilon\|_2^2]. \quad (20)$$

Only the gain network parameters ϕ are updated; the Wan backbone parameters θ remain frozen. In the first implementation, this is the only training loss. We do not add an adversarial discriminator or a separate perceptual objective. The goal is to isolate whether the gain network can learn a useful conditioning-space analysis update by itself.

5 Derivations

The conditioning-space DA update in Eq. (17) can be read as a nonlinear Kalman-style analysis step. In our TI2V setting, the DA prior is the native conditioning token itself rather than a separately predicted video. If the base conditioning token c^0 plays the role of a prior state and the observation summary y^o defines a noisy measurement, then a classical linear DA update would take the form

$$c^a = c^0 + Kr, \quad (21)$$

for some gain matrix K . Our gain network simply replaces the fixed linear gain with a learned nonlinear map:

$$Kr \rightsquigarrow G_\phi(\text{concat}(c^0, y^0, y^o, r, e_p, e_I)). \quad (22)$$

This preserves the prior–innovation–analysis structure while allowing the correction to depend on prompt semantics, anchor-image content, and observation quality.

Under the frozen diffusion model in Eq. (20), the gain network is optimized entirely through the later-half denoising objective of the frozen TI2V backbone. The current training plan therefore asks a direct question: can a single gain network learn a useful conditioning-space analysis operator while the TI2V backbone remains completely frozen?

6 Experiments and Training Plan

6.1 Datasets

Retrieval bank. We use OpenVid-1M as the external observation memory [10]. Each entry provides a caption and a real video clip, making it suitable for prompt-conditioned retrieval. In the implementation phase, we download a storage-bounded subset of the public Hugging Face mirror lodestones/OpenVid-1M and build the retrieval index only from this corpus; the retrieval bank is never used as a supervised target.

Supervised train/validation/test set. We use MSR-VTT as the supervised continuation dataset [19]. MSR-VTT contains 10K caption-annotated web videos with a standard 9K/1K train/test split. For implementation convenience, we download the public Hugging Face mirror VLM2Vec/MSR-VTT. We reserve 1K clips from the 9K training portion for validation and use the remaining 8K clips for training. Since MSR-VTT clip durations vary, we extract a canonical 10-second window per clip before splitting it into a 5-second context half and a 5-second target half.

Data separation. The retrieval bank and the supervised continuation dataset are disjoint by construction: OpenVid-1M is used only for retrieval, while MSR-VTT is used only for supervised training and evaluation. This avoids label leakage from the observation bank into the target continuation clips.

6.2 Preprocessing and Cached Features

All supervised clips are resampled to a fixed frame rate and resolution supported by the frozen Wan TI2V pipeline. For each MSR-VTT clip we store:

1. the first-half context clip x^{ctx} ,
2. the last context frame I_{last} ,
3. the ground-truth second-half target clip x^{tgt} ,
4. the native TI2V conditioning token $c^0 = C_{\text{wan}}(I_{\text{last}}, p)$,
5. the Wan target latent $z^{\text{tgt}} = E_{\text{wan}}(x^{\text{tgt}})$.

For the OpenVid retrieval bank we precompute and cache:

1. LanguageBind text and video embeddings,
2. depth summaries from Depth Anything V2,
3. motion summaries from a frozen flow extractor,
4. metadata needed to recover the original clip path and frames.

This makes the online training loop lightweight: Wan TI2V is used only to encode the native continuation condition and evaluate the frozen denoising objective, retrieval is performed over cached embeddings, and only the gain network is updated.

6.3 Training Procedure

The training procedure follows the exact prior-analysis structure in the method section.

1. Freeze Wan2.2 TI2V, LanguageBind, Depth Anything V2, and the motion extractor.
2. For each MSR-VTT training clip, keep only the last frame of the first half and the prompt as TI2V input.
3. Use the frozen Wan prompt-and-image encoder path to build the native conditioning token c^0 .
4. Build a LanguageBind retrieval query from the prompt and the last context frame.
5. Retrieve the top- K OpenVid clips and compute the weighted observation summary y^o .
6. Compute the conditioning-side summary y^0 and innovation $r = y^o - y^0$.
7. Run the gain network to obtain the assimilated conditioning token c^a .
8. Optimize only the frozen Wan denoising loss in Eq. (20).

In the first implementation, we use AdamW with a single learning rate for G_ϕ , mixed precision, and early stopping on MSR-VTT validation denoising loss. Because only one small network is trained, this loop is substantially cheaper than fine-tuning the full Wan prior.

6.4 Planned Evaluation

The primary training metric is the frozen Wan denoising loss on the MSR-VTT validation set. We also plan to decode the assimilated continuation with the frozen Wan decoder and report three secondary continuation metrics:

1. later-half reconstruction quality against the ground-truth target half,
2. depth residual between decoded continuation and target continuation,
3. transition smoothness between the last context frame and the first frames of the generated second half.

For the first project milestone, we prioritize a small number of qualitative case studies to verify that the data-assimilated token changes the later-half rollout in a meaningful way before scaling to a larger benchmark sweep.

6.5 Training Loop Pseudocode

```

for each minibatch  $(p, x^{ctx}, x^{tgt})$  from  $D_{sup}^{train}$  do
   $I_{last} \leftarrow \text{last\_frame}(x^{ctx})$ 
   $c^0 \leftarrow C_{wan}(I_{last}, p)$  // frozen Wan2.2 TI2V base condition
   $z^{tgt} \leftarrow E_{wan}(x^{tgt})$ 
   $q \leftarrow Q_{LB}(p, I_{last})$ 
   $\{(r_k, \alpha_k)\}_{k=1}^K \leftarrow \text{Retrieve}(\mathcal{R}, q)$ 
   $y^o \leftarrow \sum_{k=1}^K \alpha_k \cdot \text{ObsSummary}(r_k)$ 
   $y^0 \leftarrow H(c^0)$ 
   $r \leftarrow y^o - y^0$ 
   $\Delta c \leftarrow G_\phi(\text{concat}(c^0, y^0, y^o, r, E_{text}(p), E_{img}(I_{last})))$ 
   $c^a \leftarrow c^0 + \Delta c$ 
  sample diffusion step  $t$  and noise  $\epsilon$ 
   $z_t^{noisy} \leftarrow \alpha_t z^{tgt} + \sigma_t \epsilon$ 
   $\hat{c} \leftarrow \epsilon_\theta(z_t^{noisy}, t \mid c^a)$ 
   $\mathcal{L} \leftarrow \|\hat{c} - \epsilon\|_2^2$ 
  update  $\phi$  with AdamW; keep all other modules frozen
end for

```

7 Discussion

The main benefit of the revised formulation is cleanliness: one frozen TI2V prior, one retrieval bank, one observation projection, and one trainable gain network. The main risk is also clear: the projection from retrieved-video space into TI2V token space may discard temporal detail that would have been useful for continuation. This makes the anchor-image selection, depth summary, and motion summary the most important design choices in the whole pipeline. If the projected observation is poor, the innovation signal will be noisy even when retrieval itself is semantically correct.

Another limitation is that the current formulation still leaves the observation-to-condition projection implicit through the frozen summary operator H and the gain network. This is intentional for the first implementation because the paper aims to test whether conditioning-space DA is useful before adding more complicated architectural structure.

8 Conclusion

This paper updates RGDA into a more focused conditioning-space formulation for TI2V continuation. The central object is a data-assimilated tokenizer: a small gain network that receives the native prompt-and-image conditioning token, a retrieved observation summary, and their innovation, then outputs a corrected continuation condition for the frozen Wan DiT. We specify the observation projection needed to map retrieved videos into TI2V space, define a frozen-backbone diffusion training objective, and provide a concrete training plan using OpenVid-1M for retrieval and MSR-VTT for supervised continuation. This gives the project a clear next step: implement the cached retrieval pipeline, train the gain network, and evaluate whether conditioning-space data assimilation improves later-half video continuation without fine-tuning the full generator.

References

- [1] Caterina Buizza, César Quilodrán Casas, Philip Nadler, Julian Mack, Stefano Marrone, Zainab Titus, Clémence Le Cornec, Evelyn Heylen, Tolga Dur, Luis Baca Ruiz, et al. Data learning: Integrating data assimilation and machine learning. *Journal of Computational Science*, 58: 101525, 2022. doi: 10.1016/j.jocs.2021.101525.
- [2] Siyi Chen, Yixuan Jia, Qing Qu, He Sun, and Jeffrey A. Fessler. Flowdas: A stochastic interpolant-based framework for data assimilation. *arXiv preprint arXiv:2501.16642*, 2025. doi: 10.48550/arXiv.2501.16642.
- [3] Taiye Chen, Xun Hu, Zihan Ding, and Chi Jin. Learning world models for interactive video generation. In *Advances in Neural Information Processing Systems*, 2025. OpenReview id FzfYoUp8F1.
- [4] Yixiang Chen, Peiyan Li, Jiabing Yang, Keji He, Xiangnan Wu, Yuan Xu, Kai Wang, Jing Liu, Nianfeng Liu, Yan Huang, and Liang Wang. Bridgev2w: Bridging video generation models to embodied world models via embodiment masks. *arXiv preprint arXiv:2602.03793*, 2026. doi: 10.48550/arXiv.2602.03793.
- [5] David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018. doi: 10.48550/arXiv.1803.10122.
- [6] Muiyang He, Hanzhong Guo, Junxiong Lin, and Yizhou Yu. Video generation models as world models: Efficient paradigms, architectures and algorithms. *arXiv preprint arXiv:2603.28489*, 2026. doi: 10.48550/arXiv.2603.28489.
- [7] Daniel Hodyss and Matthias Morzfeld. Using diffusion models to do data assimilation. *arXiv preprint arXiv:2506.02249*, 2025. doi: 10.48550/arXiv.2506.02249.
- [8] Yinan Huang, Hans Hao-Hsun Hsu, Junran Wang, Bo Dai, and Pan Li. Accelerated sequential flow matching: A bayesian filtering perspective. *arXiv preprint arXiv:2602.05319*, 2026. doi: 10.48550/arXiv.2602.05319.
- [9] Peter Manshausen, Yair Cohen, Peter Harrington, Jaideep Pathak, Mike Pritchard, Piyush Garg, Morteza Mardani, Karthik Kashinath, Simon Byrne, and Noah Brenowitz. Generative data assimilation of sparse weather station observations at kilometer scales. *arXiv preprint arXiv:2406.16947*, 2024. doi: 10.48550/arXiv.2406.16947.
- [10] Kepan Nan, Rui Xie, Penghao Zhou, Tiehan Fan, Zhenheng Yang, Zhijie Chen, Xiang Li, Jian Yang, and Ying Tai. Openvid-1m: A large-scale high-quality dataset for text-to-video generation. In *The Thirteenth International Conference on Learning Representations*, 2025. OpenReview id j7kdXSrISM.
- [11] Yuta Oshima, Masahiro Suzuki, Yusuke Iwasawa, Yutaka Matsuo, and Hiroki Furuta. World-pack: Compressed memory improves spatial consistency in video world modeling. *arXiv preprint*, 2025. OpenReview id CuNH3zxgm.
- [12] Elia Peruzzo, Dejia Xu, Xingqian Xu, Humphrey Shi, and Nicu Sebe. Ragme: Retrieval augmented video generation for enhanced motion realism. *arXiv preprint arXiv:2504.06672*, 2025. doi: 10.48550/arXiv.2504.06672.
- [13] Jonas Scholz and Richard E. Turner. Warm starts accelerate conditional diffusion. *arXiv preprint arXiv:2507.09212*, 2025. doi: 10.48550/arXiv.2507.09212.
- [14] Yu Shang, Lei Jin, Yiding Ma, Xin Zhang, Chen Gao, Wei Wu, and Yong Li. Longscape: Advancing long-horizon embodied world models with context-aware moe. *arXiv preprint arXiv:2509.21790*, 2025. doi: 10.48550/arXiv.2509.21790.
- [15] Wan Team. Wan2.2. GitHub repository, 2026. URL <https://github.com/Wan-Video/Wan2.2>.

- [16] Jiahao Wang, Luoxin Ye, Taiming Lu, Junfei Xiao, Jiahan Zhang, Yuxiang Guo, Xijun Liu, Rama Chellappa, Cheng Peng, Alan Yuille, and Jieneng Chen. Evoworld: Evolving panoramic world generation with explicit 3d memory. *arXiv preprint arXiv:2510.01183*, 2025. doi: 10.48550/arXiv.2510.01183.
- [17] Luozhou Wang, Zhifei Chen, Yihua Du, Dongyu Yan, Wenhao Ge, Guibao Shen, Xinli Xu, Leyi Wu, Man Chen, Tianshuo Xu, Peiran Ren, Xin Tao, Pengfei Wan, and Ying-Cong Chen. A mechanistic view on video generation as world models: State and dynamics. *arXiv preprint arXiv:2601.17067*, 2026. doi: 10.48550/arXiv.2601.17067.
- [18] Fan Wu, Jiacheng Wei, Ruibo Li, Yi Xu, Junyou Li, Deheng Ye, and Guosheng Lin. Ic-world: In-context generation for shared world modeling. *arXiv preprint arXiv:2512.02793*, 2025. doi: 10.48550/arXiv.2512.02793.
- [19] Jun Xu, Tao Mei, Ting Yao, and Yong Rui. Msr-vtt: A large video description dataset for bridging video and language. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 5288–5296, 2016. doi: 10.48550/arXiv.1512.08895.
- [20] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything v2. *arXiv preprint arXiv:2406.09414*, 2024. doi: 10.48550/arXiv.2406.09414.
- [21] Jiahan Zhang, Muqing Jiang, Nanru Dai, Taiming Lu, Arda Uzunoglu, Shunchi Zhang, Yana Wei, Jiahao Wang, Vishal M. Patel, Paul Pu Liang, Daniel Khashabi, Cheng Peng, Rama Chellappa, Tianmin Shu, Yilun Du, and Jieneng Chen. World-in-world: World models in a closed-loop world. *arXiv preprint arXiv:2510.18135*, 2025. doi: 10.48550/arXiv.2510.18135.
- [22] Hongkai Zheng, Wenda Chu, Austin Wang, Nikola Kovachki, Ricardo Baptista, and Yisong Yue. Ensemble kalman diffusion guidance: A derivative-free method for inverse problems. *Transactions on Machine Learning Research*, 2025. doi: 10.48550/arXiv.2409.20175. arXiv:2409.20175.
- [23] Siyuan Zhou, Yilun Du, Yuncong Yang, Lei Han, Peihao Chen, Dit-Yan Yeung, and Chuhan Gan. Learning 3d persistent embodied world models. In *Advances in Neural Information Processing Systems*, 2025. OpenReview id XTTbzC7O2T.
- [24] Bin Zhu, Bin Lin, Munan Ning, Yang Yan, Jiayi Cui, HongFa Wang, Yatian Pang, Wenhao Jiang, Junwu Zhang, Zongwei Li, Wancai Zhang, Zhifeng Li, Wei Liu, and Li Yuan. Languagebind: Extending video-language pretraining to n-modality by language-based semantic alignment. *arXiv preprint arXiv:2310.01852*, 2024. doi: 10.48550/arXiv.2310.01852.